

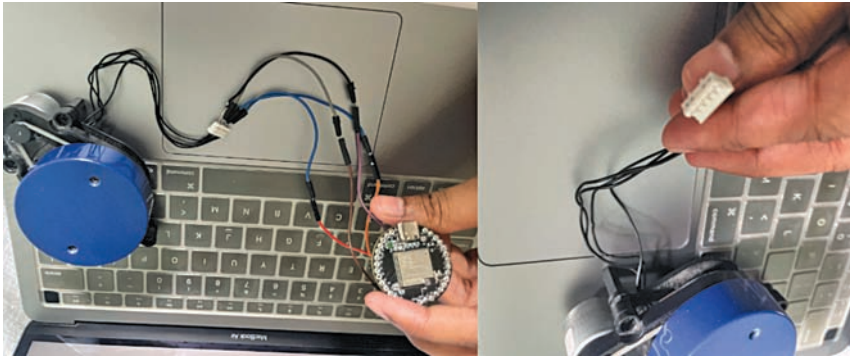


Fig. 7: Lidar pin connection and lidar connector



Fig. 8: Lidar pin USB connector module marked with pin numbers

battery, with the lidar controlled via PWM signals through the RX pin. Output is sent to the serial port.

Software

Depending on the lidar version, you can use the appropriate Arduino library. Here, a library compatible with the YD X2 lidar to retrieve data was utilised.

First, configure the Wi-Fi SSID and password in the code. You can connect the lidar to an existing Wi-Fi network or set it as a Wi-Fi access point (AP). In this project, we set the lidar as an AP. Fig. 5 shows the code snippet for configuring the AP.

Next, define the serial pins, ports, and PWM motor control pin in the code. Write the web page using HTML,

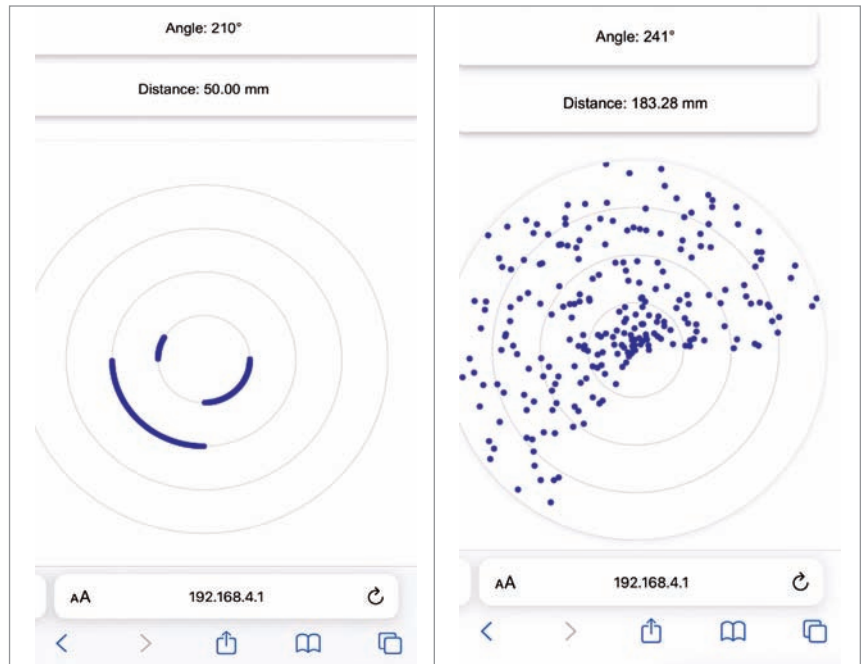


Fig. 9: Lidar scan data

CSS, and JavaScript to display the map and data in real time over Wi-Fi. Fig. 6 shows the web page's code snippet.

Construction and testing

After uploading the source code, complete the wiring as per Fig. 4. Connect the lidar motor control pin to any PWM-enabled pin on the IndusBoard. The serial data pin can also be connected to any pin, but in this case, RX pin 43 was used.

For testing, connect the lidar's connector either via a header pin or by soldering directly to the board. Mount the IndusBoard and lidar as shown in

EFY Note

The source code of this project is available for free download at www.electronicsforu.com

Fig. 7. The IndusBoard, being small enough, fits behind the lidar. Fig. 8 shows the lidar pin USB connector module, with pin numbers marked.

After assembly, power on the lidar and establish a Wi-Fi connection. Open a web browser and navigate to the IndusBoard's default IP address: 192.168.4.1. The web page will display the real-time map and provide distance and angle data. Fig. 9 shows the lidar scan data plotted on the web page. **EFY**

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